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ASSIGNMENT-1

Subject : Computer Networks

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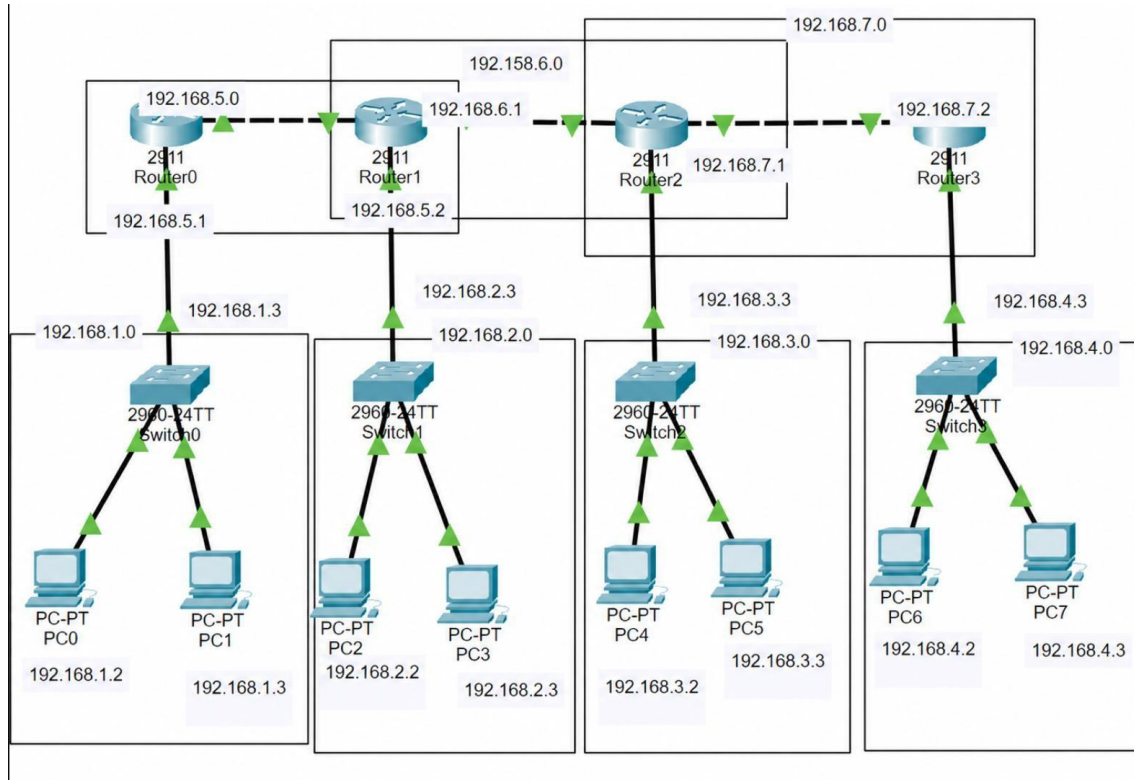
Submitted to:

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Q1. Static Routing with 4 Routers, 4 Switches & 8 PCs

To demonstrate static routing, a network was designed in Cisco Packet Tracer consisting of four routers (R1, R2, R3, R4), four switches (SW1-SW4) and eight PCs (two PCs per switch, PC1-PC8).

Network Topology Diagram



Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC5	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC0	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC7	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC5	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC1	PC7	ICMP		0.000	N	1	(edit)	(delete)

With these static routes in place, every PC (PC1 through PC8) can successfully ping every other PC across the network, even though they lie in different LANs separated by multiple routers.

Q2. Difference between Hub, Switch & Router

<i>Parameter</i>	<i>Hub</i>	<i>Switch</i>	<i>Router</i>
<i>OSI Layer</i>	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
<i>Working Principle</i>	Broadcasts incoming data to all ports	Forwards data only to the destination port using MAC address	Forwards data between different networks using IP address
<i>Addressing Used</i>	None	MAC Address	IP Address
<i>Collision Domain</i>	Single collision domain for all ports	Separate collision domain per port	Separate collision domain per port
<i>Broadcast Domain</i>	Single broadcast domain	Single broadcast domain (unless VLANs used)	Separate broadcast domain per interface
<i>Intelligence</i>	No intelligence, purely electrical repeater	Learns MAC addresses, moderately intelligent	Uses routing tables/protocols, most intelligent
<i>Data Format</i>	Bits (electrical signal)	Frames	Packets
<i>Typical Use</i>	Connecting a few devices in a very small, legacy LAN	Connecting devices within the same LAN	Connecting different networks (LAN to LAN, LAN to WAN/Internet)

Why is the Router the most intelligent device?

A router operates at the Network Layer and takes forwarding decisions based on logical IP addressing rather than just electrical signals or hardware addresses. It maintains a routing table, can run dynamic routing protocols (RIP, OSPF, EIGRP) to learn about remote networks, performs path selection when multiple routes exist, can filter traffic using access control lists, and connects entirely different networks together (including the Internet). This ability to understand addressing schemes, make intelligent path decisions, and interconnect dissimilar networks is why the router is considered the most intelligent networking device.

Why is the Hub the least intelligent device?

A hub is a purely physical-layer device. It has no knowledge of MAC addresses or IP addresses and cannot distinguish one connected device from another. Whenever it receives a signal on one port, it simply regenerates and broadcasts that signal out of every other port, regardless of the intended destination. It cannot filter, switch, or route traffic in any way, which also means every device connected to a hub shares the same collision domain. Because it performs no processing or decision-making at all, the hub is considered the least intelligent device in the hierarchy of Hub, Switch and Router.

Q3. Random Access Protocols

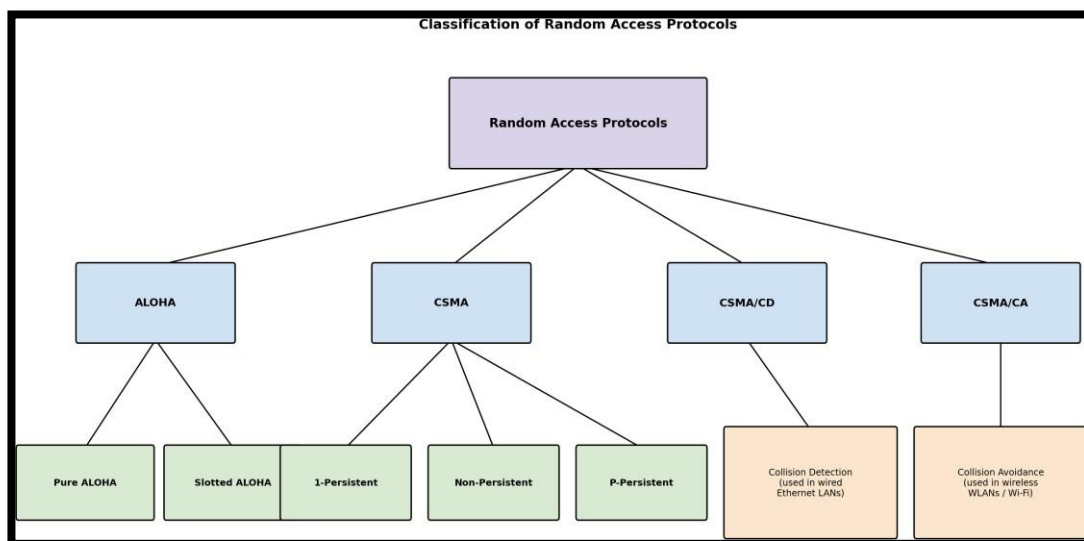
What are Random Access Protocols?

Random Access Protocols are a category of Multiple Access Control protocols used in the Data Link Layer (MAC sub-layer) in which no station is given a fixed turn to transmit and there is no central controller managing access to the shared communication channel. Every station is free to send data whenever it has something to transmit, and it decides on its own when to send. Because multiple stations can attempt to transmit at the same time, there is always a possibility of collision, so these protocols also define rules for how a station should behave to detect, avoid, or recover from such collisions.

Why do we need Random Access Protocols?

In a shared broadcast medium such as a LAN or a wireless channel, several stations must be able to communicate using the same channel. Fixed-turn based access methods (like TDMA or polling) work well for predictable, uniform traffic but waste bandwidth when traffic is bursty or unpredictable, since a station's turn goes unused if it has nothing to send. Random access protocols solve this problem by allowing any station to access the medium on demand, which gives much better channel utilization for the bursty, random traffic patterns typical of computer networks. They are simple to implement, do not require a central coordinator, and adapt naturally to varying numbers of active stations, which is why they are widely used in LAN technologies such as Ethernet and Wi-Fi.

Classification Diagram



1. ALOHA

ALOHA was one of the earliest random access protocols, developed at the University of Hawaii. A station transmits whenever it has data, with no coordination with other stations.

- **Pure ALOHA:** A station transmits data as soon as it is ready, without checking whether the channel is free. If two frames collide, both are destroyed, and each sender waits a random amount of time before retransmitting. Because transmissions can start at any instant, the

vulnerable time for collision is $2 \times$ frame transmission time, giving a maximum channel utilization of about 18%.

- Slotted ALOHA: Time is divided into fixed-length slots equal to the frame transmission time, and a station is allowed to transmit only at the beginning of a slot. This reduces the vulnerable time to just one frame time, doubling the maximum efficiency to about 37%.

2. CSMA (Carrier Sense Multiple Access)

CSMA improves on ALOHA by requiring a station to first sense the carrier (listen to the channel) before transmitting, which reduces the chance of collision. Depending on what a station does after sensing the channel, CSMA has three variations:

- 1-Persistent CSMA: If the channel is busy, the station keeps sensing continuously and transmits immediately (with probability 1) as soon as the channel becomes idle. This is simple but can cause collisions if multiple stations are waiting.
- Non-Persistent CSMA: If the channel is busy, the station does not keep sensing continuously; instead it waits for a random amount of time and then senses again. This reduces collisions but increases delay.
- P-Persistent CSMA: Used mainly in slotted channels. If the channel is idle, the station transmits with probability p , and with probability $(1-p)$ it defers to the next slot; this balances the trade-off between the aggressiveness of 1-persistent and the delay of non-persistent CSMA.

3. CSMA/CD (Collision Detection)

CSMA/CD adds collision detection to CSMA: a transmitting station continues to monitor the channel while sending. If it detects that its signal is colliding with another station's signal, it immediately stops transmitting, sends a short jam signal to inform all stations of the collision, and then retransmits after a random backoff period (typically using the binary exponential backoff algorithm). CSMA/CD is used in traditional wired Ethernet networks.

4. CSMA/CA (Collision Avoidance)

CSMA/CA is designed for wireless networks, where a station cannot easily detect a collision while transmitting (since it cannot listen and send at the same time, and due to the hidden terminal problem). Instead, it focuses on avoiding collisions: a station senses the channel, and if idle, waits for an Inter-Frame Space plus a random backoff before transmitting; optionally, RTS (Request to Send) and CTS (Clear to Send) control frames are exchanged to reserve the channel before the actual data is sent. CSMA/CA is used in wireless LANs such as Wi-Fi (IEEE 802.11).

Difference between CSMA/CD and CSMA/CA

<i>Parameter</i>	<i>CSMA/CD</i>	<i>CSMA/CA</i>
<i>Full Form</i>	Carrier Sense Multiple Access with Collision Detection	<i>Carrier Sense Multiple Access with Collision Avoidance</i>
<i>Approach</i>	<i>Detects a collision after it occurs and then retransmits</i>	<i>Tries to avoid a collision before it happens</i>
Medium Used	Wired networks (traditional Ethernet, IEEE 802.3)	Wireless networks (Wi-Fi, IEEE 802.11)
Mechanism	Station monitors the medium while transmitting; on detecting a collision it stops, sends a jam signal, and retransmits after a random backoff	Station senses the channel, waits for it to be idle, then waits an additional random backoff/IFS time and may use RTS/CTS control frames before sending
Reason	Collisions are easy to detect on a wired medium since signal levels can be monitored	Collisions are hard to detect on a wireless medium (hidden terminal problem, cannot transmit and listen at the same time), so avoidance is used instead
Efficiency	Simpler and faster since it reacts only when a collision occurs	Comparatively slower due to extra waiting time and control overhead used to avoid collisions